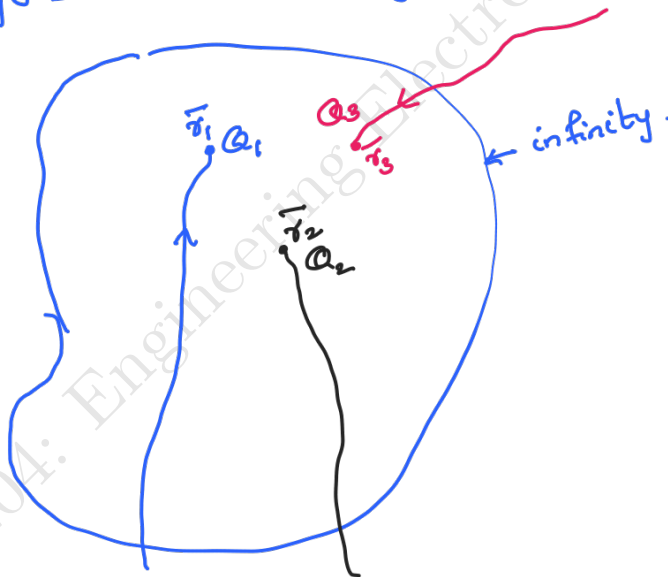


Energy in Electric field.

Q.



$\Rightarrow$ Workdone in bringing charge 'q' from $\vec{r}_0$ to $\vec{r}_1$ is;
 Workdone = $\int_{\vec{r}_0}^{\vec{r}_1} \vec{E} \cdot d\vec{l} = q\phi(\vec{r}_0) - q\phi(\vec{r}_1)$
 $=$ Kinetic energy gained by charge 'q'.

Energy required to bring 'N' charges;

Workdone in bringing Q_1 to $\vec{r}_1 = 0$.

$$\Rightarrow W_1 = 0.$$

Workdone in bringing Q_2 to $\vec{r}_2$

$$W_2 = Q_2 \phi(\vec{r}_2)$$

$$= Q_2 \frac{Q_1}{4\pi\epsilon_0 |\vec{r}_2 - \vec{r}_1|}$$

Workdone in bringing Q_3 to $\vec{r}_3$.

$$W_3 = Q_3 \phi(\vec{r}_3)$$

$$= Q_3 \left[\frac{Q_1}{4\pi\epsilon_0 |\vec{r}_3 - \vec{r}_1|} + \frac{Q_2}{4\pi\epsilon_0 |\vec{r}_3 - \vec{r}_2|} \right]$$

$\vdots$
 W_N



Total workdone in bringing 'N' charges;

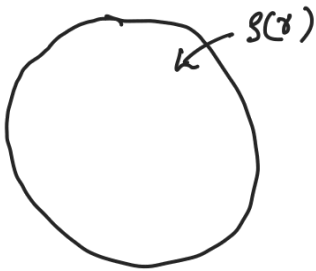
$$W = W_1 + W_2 + \dots + W_N.$$

$$W = \sum_{i=1}^N Q_i \sum_{j=1}^{i-1} \frac{Q_j}{4\pi\epsilon_0 |\vec{r}_i - \vec{r}_j|}$$

← But this expression is not so obvious to calculate.

For better understanding of this problem;

Let us try to find the energy required to build up any charge distribution $\rho(r)$



Keeping the shape same, let us try to build uniform charge distribution $\rho(r)$ gradually over time.

⇒ Say at $t=0 \rightarrow$ no charge
 $t=t_0 \rightarrow$ bringing $\epsilon \rho(r)$ charges.
 (say 0.01% of $\rho(r)$ i.e; $\epsilon \ll \rho$)
 $t=2t_0 \rightarrow 2\epsilon \rho(r)$
 $t=3t_0 \rightarrow 3\epsilon \rho(r)$
 $\vdots$
 $t=Nt_0 \rightarrow N\epsilon \rho(r) = \rho(r)$
 $\Rightarrow \epsilon = 1/N$

At $t=kt_0$, charge build is $k\epsilon \rho(r)$
 $= \frac{k}{N} \rho(r)$

$$\begin{aligned}
 \text{At } t = kt_0, \quad \phi &= \frac{1}{4\pi\epsilon_0} \iiint_{\text{Vol.}} \frac{\frac{k}{N} \rho(\vec{r}')}{|\vec{r} - \vec{r}'|} dV' \\
 &= \frac{k}{N} \underbrace{\frac{1}{4\pi\epsilon_0} \iiint_{\text{Vol.}} \frac{\rho(\vec{r}')}{|\vec{r} - \vec{r}'|} dV'}_{\phi(\vec{r})}
 \end{aligned}$$

ie; At $t = kt_0$, $\phi = \frac{k}{N} \phi(\vec{r})$

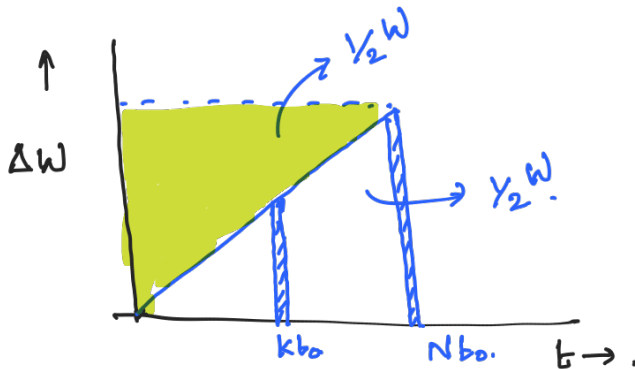
Now at $t = (k+1)t_0$ to bring $\frac{\rho(\vec{r})}{N}$ charge into the existing charge distribution $\frac{k}{N} \rho(\vec{r})$; the additional work done is:

$$\begin{aligned}
 \Delta W &= \text{charge} \times \text{potential.} \\
 &= \iiint_{\text{Vol.}} \frac{\rho(\vec{r})}{N} dV \cdot \frac{k}{N} \phi(\vec{r})
 \end{aligned}$$

Total work done or energy required for building $\rho(\vec{r})$ is;

$$\begin{aligned}
 W &= \sum_{i=1}^N \Delta W_i = \left[\iiint_{\text{Vol.}} \frac{\rho(\vec{r})}{N^2} \phi(\vec{r}) dV \right] \sum_{i=1}^N i \\
 &\quad \frac{N(N+1)}{2} \approx \frac{N^2}{2} \\
 &= \iiint_{\text{Vol.}} \frac{\rho(\vec{r})}{N^2} \phi(\vec{r}) \frac{N^2}{2} dV.
 \end{aligned}$$

$$\boxed{W = \frac{1}{2} \iiint_{\text{Vol.}} \rho(\vec{r}) \phi(\vec{r}) dV.}$$

Intuitive explanation:

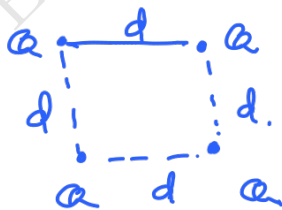
Work required is triangle.

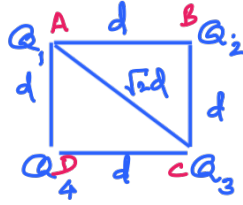
- * Initially No work.
- * $\frac{1}{2}$ way $\rightarrow \frac{1}{2}$ work.
- * Finally we do full work.
- $\therefore$ Total work = Area of Triangle shown.
- $\rightarrow \frac{1}{2}$ Comes from area of this triangle.

* The total workdone would have been

$\rightarrow$ area of rectangle if from the beginning onwards hypothetically the workdone to build the charge is against the potential $\phi(\vec{r})$ due to full charge density (instead of at kb_0 ; $\phi = \frac{k}{N} \phi(\vec{r})$)

Example: Calculate the energy required to build the following charge distribution:



Solution:

$$\phi_A = \frac{Q_2}{4\pi\epsilon_0 d} + \frac{Q_3}{4\pi\epsilon_0 (\sqrt{2}d)} + \frac{Q_4}{4\pi\epsilon_0 d}.$$

$$\text{Here } Q_1 = Q_2 = Q_3 = Q_4 = Q$$

$$\Rightarrow \phi_A = \frac{Q}{4\pi\epsilon_0} \left[\frac{2}{d} + \frac{1}{\sqrt{2}d} \right]$$

$$= \frac{Q}{4\pi\epsilon_0} \left(\frac{2\sqrt{2} + 1}{\sqrt{2}d} \right)$$

$$\text{Here } \phi_A = \phi_B = \phi_C = \phi_D = \phi = \frac{Q}{4\pi\epsilon_0} \left(\frac{2\sqrt{2} + 1}{\sqrt{2}d} \right) \text{ J/C}$$

$$\text{Energy required} = \text{Work done} = \frac{(4 Q \phi)}{2}$$

$$= 2Q\phi = \frac{Q^2 (4 + \sqrt{2})}{4\pi\epsilon_0 d}.$$

Sanity check

$$W = 0 + Q \left[\frac{Q}{4\pi\epsilon_0} \cdot \frac{1}{d} \right] + Q \left[\frac{Q}{4\pi\epsilon_0} \frac{1}{d} + \frac{Q}{4\pi\epsilon_0 2\sqrt{2}d} \right]$$

$$+ Q \left[\frac{Q}{4\pi\epsilon_0 d} + \frac{Q}{4\pi\epsilon_0 \sqrt{2}d} + \frac{Q}{4\pi\epsilon_0 d} \right]$$

$$W = Q \left[4 \frac{Q}{4\pi\epsilon_0} \cdot \frac{1}{d} + 2 \frac{Q}{4\pi\epsilon_0 \sqrt{2}d} \right]$$

$$= \frac{Q^2}{4\pi\epsilon_0 d} \left[4 + \frac{2}{\sqrt{2}} \right] \checkmark$$

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